Industrial Safety Processes

Confined Spaces

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Confined Space – CLC II (11.1)

confined space means an enclosed or partially enclosed space that

a) is not designed or intended for human occupancy except for the purpose of performing work,

b) has restricted means of access and egress, and may become hazardous to any person entering it owing to

   i. its design, construction, location or atmosphere,

   ii. the materials or substances in it, or

   iii. any other conditions relating to it;

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Class of Confined Spaces – CLC II (11.1)

class of confined spaces means a group of at least two confined spaces that are likely, by reason of their similarity, to present the same hazards to persons entering, exiting or occupying them;

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Hot Work – CLC II (11.1)

hot work means any work where flame is used or a source of ignition may be produced.

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Qualified Person – CLC II (1.2)

qualified person means, in respect of a specified duty, a person who, because of his knowledge, training and experience, is qualified to perform that duty safely and properly

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Comparing to Other Provinces

- Definitions are similar, but not exact
- Some provinces provide lesser provisions for “confined spaces” that do not represent significant hazards other than their construction (that they are a confined space)
- This can cause confusion when moving between jurisdictions

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British Columbia

"confined space", except as otherwise determined by the Board, means an area, other than an underground working, that

a) is enclosed or partially enclosed,

b) is not designed or intended for **continuous human occupancy**,

c) has limited or restricted means for entry or exit that may complicate the provision of first aid, evacuation, rescue or other emergency response service, and

d) is large enough and so configured that a worker could enter to perform assigned work;

British Columbia OH&S Regulation, 9.1 Definitions

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Alberta

"confined space" means a restricted space **which may become hazardous to a worker** entering it because of

(a) an atmosphere that is or may be injurious by reason of oxygen deficiency or enrichment, flammability, explosivity, or toxicity,

(b) a condition or changing set of circumstances within the space that presents a potential for injury or illness, or

(c) the potential or inherent characteristics of an activity which can produce adverse or harmful consequences within the space;

Alberta OH&S Code, Part 1

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Saskatchewan

(a) "confined space" means an enclosed or partially enclosed space that:

i. is **not primarily designed or intended for human occupancy, except for the purpose of performing work**; and

ii. has restricted means of entrance and exit;

Saskatchewan OH&S Regulation, Section 266

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Saskatchewan

(b) "**hazardous confined space**" means a confined space that is or may become hazardous to a worker entering the confined space due to:

i. the design, construction or atmosphere of the confined space;

ii. the materials or substances in the confined space;

iii. the work activities or processes used in the confined space; or

iv. any other conditions relating to the confined space;

Saskatchewan OH&S Regulation, *Section 266*

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Takeaways

- Three main elements are common across all definitions:
  - Enclosed or partially enclosed
  - Not intended for human occupancy
  - Limited means of entry / exit
- Not all definitions of a “confined space” include hazards other than the confined space itself (BC, Saskatchewan)

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Identification

- Should be initially identified through a formal hazard assessment process
- Regulators prefer job safety analysis (JSAs) as it will most likely capture things like entering these kinds of spaces
- Canada Labour Code Part II requires assessments of all confined spaces (11.2)

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Labeling / Signage

- Most jurisdictions require some form of labelling or signage for confined spaces in the workplace
- This often includes notification of workplace requirements – permit required, authorized entry only, etc.

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Securing

- Most jurisdictions also require that confined spaces be "controlled" to prevent unauthorized access
- This could include bolted hatches, locks, or the requirement to enter through other controlled areas to access the confined space
- Usually signage is not considered sufficient for potentially IDLH confined space situations

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Permitting

- Most jurisdictions require a permitting system to authorize confined space entry
- Permits usually include information on known or expected hazards, assessments, special controls, rescue plans, and a location to record atmospheric tests
- Permits are usually required to be kept at the entry location for the confined space

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Non-Permitted Spaces

- Some “confined spaces” may not require permits based on local provincial regulation
- In Alberta these are “restricted spaces” and are generally known as “no-permit confined spaces”
- This is limited to situations were hazards are basically the space itself and nothing else

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Hazard Assessments – CLC II (11.2)

11.2 (1) If a person could be required to enter a confined space in order to perform work for an employer and an assessment under this subsection has not been carried out in respect of the confined space or in respect of the class of confined spaces to which it belongs, the employer shall appoint a qualified person

a) to carry out an assessment of the physical and chemical hazards to which the person is likely to be exposed in the confined space or the class of confined spaces; and

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Hazard Assessments – CLC II (11.2)

b) to specify the tests that are necessary to determine whether the person would be likely to be exposed to any of the hazards identified pursuant to paragraph (a).

2) The qualified person referred to in subsection (1) shall, in a signed and dated report to the employer, record the findings of the assessment carried out pursuant to paragraph (1)(a).

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Typical Confined Space Issues

- Atmospheric problems
  - Oxygen deprivation or enrichment
  - Toxic or explosive gases, chemical vapours/fumes
- Changing circumstances
  - Automatic start up of machinery
  - Change in temperatures, lighting, etc.

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Typical Confined Space Issues

- Harmful consequences of work performed
  - Fumes / sparks / slag from hot work
  - Working at heights, working over water
- Other Issues
  - Equipment working nearby, vessel movement, weather
  - Animals, feces, insects, moulds

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Training – CLC II (11.11)

11.11 (1) The employer shall provide every employee who is likely to enter a confined space with instruction and training in

a) the procedures established pursuant to paragraphs 11.3(a) and 11.5(1)(a); and

b) the use of the protection equipment referred to in paragraphs 11.3(b), (c) and (d).

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Typical Training

- Applicable legislative requirements
- Typical hazard and permitting
- Atmospheric testing (basic)
- SCBA / SABA use (not always)
- Basic confined space entry and rescue practice (not always)

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Attendant Training

- Procedures for entry, including permitting & rescue
  - But is not permitted to engage in rescue
  - 6 out of 10
- Alarm or communication system
- Operation of gas monitoring and ventilation systems

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Rescue Training

- Variable techniques for access and retrieval
- Use of various respiratory protection devices
- Use of various rescue equipment:
  - Stretchers and body boards
  - Tripods and winches
  - High-angle and rope rescue techniques

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Drills and Recerts

- Re-training is usually conducted every 3 years
- Drills for rescue should be conducted at least annually, but recommend more often if:
  - A lot of confined space work is done,
  - There are significant IDLH hazards, or
  - The workplace is remote

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Entry Procedures – CLC II (11.3)

- Created in consultation with the local health and safety committee or representative, including information to be included on confined space entry permits
- But employer responsible for specifying controls to be used in conjunction of report prepared by qualified person

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Entry Management – CLC II (11.4)

- Entry report to be prepared by a qualified person which includes:
  - Verification of atmospheric testing requirements
  - Controls in place for liquid, moving solids, mechanical, and other hazards (lockout, double block and bleed, etc.)
  - Safe entry possible by entrants with all required PPE and equipment
  - Required emergency response equipment

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Hot Work – CLC II (11.9)

11.9 (1) Unless a qualified person has determined that the work can be performed safely, hot work shall not be performed in a confined space that contains

a) an explosive or flammable hazardous substance in a concentration in excess of 10 per cent of its lower explosive limit; or

b) oxygen in a concentration in excess of 23 per cent.

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Hot Work – CLC II (11.9)

- Additional precautions are required if there a risk of creating a fire due to flammable / combustible substances
- Also need to consider potential of hot work to create additional hazards (fumes, heat, oxygen displacement)
- Welding in confined spaces can be very hazardous due to the introduction of welding gasses

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Atmospheric Testing

- The testing performed will be based on the expected contaminants in the confined space
- These are usually classified as known or “last-known”, suspected, or unknown
- Will also include any impacts due to the nature of the work being performed
  - Welding, cutting, coating, cleaning, etc.

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Atmospheric Testing

- Snorkel systems are used to permit persons outside a space to sample air inside the space
- Entrants should also use personal gas monitors
- Really important to understand how gases layer in a space and good practice in sampling the entire atmosphere

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Oxygen Levels in Confined Spaces

Jurisdiction	Lower O_2 Limit	Upper O_2 Limit
Federal	**18**	23
BC	**19.5**	23
Alberta	**19.5**	23
Saskatchewan	**19.5**	23
Ontario	**19.5**	23

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Gas Layering

- Different gasses will settle at different levels in a confined space, based on a combination of:
  - Their weight relative to air
  - Temperature
  - Agitation (air flow or stirring up liquid / solid media)

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Ventilation Equipment – CLC II (11.10)

- Required to maintain airborne contaminants to acceptable levels or ensure proper oxygen levels
- Monitoring and alarm systems are required for the equipment

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Harness – CLC II (11.5)

- In situations where there is an identified or suspected atmospheric hazard or chemical agents, entrants must wear “an appropriate safety harness that is securely attached to a lifeline” that is:
  - “…attached to a secure anchor outside the confined space”
  - Is managed be the attendant
  - Does not create additional hazards
  - Equipped with a mechanical retrieval device if feasible

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